Supplemental Material

Data S1.

Supplemental Methods

The applied search terms

Search terms applied with MEDLINE/PUBMED:

The used search terms: “colchicine“, “colgout”, “colcemid“, “beta-lumicolchicine” “colchiquim”, “colchisol”, “colchicum”, “colchicin”, ”colchicinum”, “colchizin” AND (All-cause mortality OR Cardiovascular mortality OR Myocardial infarction fatal and non-fatal OR adverse events OR Acute Coronary Syndrom OR coronary artery disease OR CAD OR Stroke fatal and non-fatal OR cardiovascular revascularization OR Non-scheduled cardiovascular interventions OR Non-scheduled hospitalizations OR cardiovascular disease OR STEMI OR NSTEMI or percutaneous coronary intervention OR cardiac arrest.

Table S1. Risk of bias assessment.

Study	Year	Follow-up duration	Random sequence generation	Allocation concealment	Blinding of participant/ personnel	Blinding of outcome assessment	Incomplete outcome data	Selective reporting	Intention to treat analysis	<5% patients with missing outcome data
O´Keefe et al.[28]	1992	6 months	L	L	L	L	L	L	Yes	Yes
COOL trial[29]	2011	30 days	L	L	L	L	L	L	Yes	No
LoDoCo trial[30]	2013	36 months	L	H	H	L	L	L	Yes	Yes
Deftereos et al.[31]	2013	6 months	L	L	L	L	L	L	Yes	No
Giannopoulos et al.[32]	2015	10 days	L	L	L	L	L	L	Yes	Yes
Deftereos et al.[33]	2015	5 days	L	L	L	L	L	L	Yes	Yes
Zarpelon et al.[34]	2016	14 days	L	H	H	H	L	L	Unclear	Yes
COLIN trial[35]	2016	1 month	L	H	H	H	L	L	Yes	Yes
LoDoCo-MI trial[36]	2019	30 days	L	L	L	L	L	L	Yes	Yes
COLCOT trial[8]	2019	22.6 months	L	L	L	L	L	L	Yes	Yes
COLCHICINE-PCI trial[16]	2020	30 days	L	L	L	L	L	L	Yes	Yes
COPS trial[17]	2020	400 days	L	L	L	L	L	L	Yes	Yes
LoDoCo2 trial[18]	2020	(median) 28.6 months	L	L	L	L	L	L	Yes	Yes

H = high risk; L = low risk; U = unclear risk.

Table S2. Assessment of the studies´ quality according to the *GRADE* (Grading of Recommendations, Assessment, Development and Evaluations) tool

Certainty assessment							№ of patients		Effect		Certainty	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Colchicine	placebo / control	Relative (95% CI)	Absolute (95% CI)		
All cause mortality in CAD patients												
13	randomized trials	not serious [a]	not serious	not serious	serious [b]	none	136/6602 (2.1%)	127/6496 (2.0%)	**OR 0.96** (0.65 to 1.41)	**1 fewer per 1,000** (from 7 fewer to 8 more)	⊕⊕⊕◯ MODERATE	CRITICAL
Cardiovascular (CV) mortality in CAD patients												
9	randomized trials	not serious	not serious	not serious	not serious	none	45/6165 (0.7%)	57/6137 (0.9%)	**OR 0.82** (0.55 to 1.22)	**2 fewer per 1,000** (from 4 fewer to 2 more)	⊕⊕⊕⊕ HIGH	CRITICAL
Stroke or TIA												
7	randomized trials	not serious	serious [c]	not serious	serious [b]	none	26/6312 (0.4%)	54/6272 (0.9%)	**OR 0.50** (0.31 to 0.81)	**4 fewer per 1,000** (from 6 fewer to 2 fewer)	⊕⊕◯◯ LOW	CRITICAL
New myocardial infarction (MI)												
7	randomized trials	not serious	serious [d]	not serious	serious [b,e]	none	192/6154 (3.1%)	263/6121 (4.3%)	**OR 0.64** (0.46 to 0.90)	**15 fewer per 1,000** (from 23 fewer to 4 fewer)	⊕⊕◯◯ LOW	CRITICAL
Ischemia-driven revascularization												
5	randomized trials	not serious	serious [f]	not serious	not serious [e]	none	167/5842 (2.9%)	244/5842 (4.2%)	**OR 0.61** (0.42 to 0.88)	**16 fewer per 1,000** (from 24 fewer to 5 fewer)	⊕⊕⊕◯ MODERATE	IMPORTANT
Gastrointestinal side effects												

Certainty assessment							№ of patients		Effect		Certainty	Importance
№ of studies	Study design	Risk of bias	Inconsistency	Indirectness	Imprecision	Other considerations	Colchicine	placebo / control	Relative (95% CI)	Absolute (95% CI)		
12	randomized trials	not serious	not serious	not serious	serious [b,e]	none	734/6487 (11.3%)	588/6389 (9.2%)	**OR 2.21** (1.45 to 3.36)	**91 more per 1,000** (from 36 more to 162 more)	⨁⨁⨁◯ MODERATE	IMPORTANT
Infectious complications												
3	randomized trials	not serious	serious [g]	not serious	serious [e]	none	207/5163 (4.0%)	188/5175 (3.6%)	**OR 1.42** (0.81 to 2.47)	**14 more per 1,000** (from 7 fewer to 49 more)	⨁⨁◯◯ LOW	IMPORTANT
Therapy discontinuation / withdrawal rate												
11	randomized trials	not serious	not serious	not serious	serious [b,e]	none	906/6313 (14.4%)	789/6226 (12.7%)	**OR 1.68** (1.14 to 2.48)	**69 more per 1,000** (from 15 more to 138 more)	⨁⨁⨁◯ MODERATE	IMPORTANT
Non-cardiovascular mortality												
4	randomized trials	not serious	serious [g]	not serious	not serious	none	85/5806 (1.5%)	60/5788 (1.0%)	**OR 1.35** (0.90 to 2.02)	**4 more per 1,000** (from 1 fewer to 10 more)	⨁⨁⨁◯ MODERATE	CRITICAL

95%CI = 95% Confidence interval; CAD = Coronary artery disease; OR: Odds ratio; TIA = Transient ischemic attack.

* *Grading of Recommendations, Assessment, Development and Evaluation (GRADE) Working Group (Handbook for grading the quality of evidence and the strength of recommendations using the GRADE approach. Updated October 2013.)*

Explanations:

a. 3 out of 13 randomized studies had an open-label design, which is prone to some bias.
b. Wide confidence intervals (CI), which were observed among some of the included studies, may lower the quality of evidence.
c. Definition of transient ischemic attack (TIA) or stroke varied across the included studies.
d. Definition of myocardial infarction (MI) differed among the included studies.
e. There was a considerable heterogeneity (I2 >25%) among the included studies.
f. Definition of ischemia driven revascularization (IDR) varied among the included studies.
g. This adverse outcome has not been systematically reported by all trials.

Table S3. Applied definitions of myocardial infarction and stroke per the individual studies.

Study	Year	Myocardial infarction*	Stroke
O´Keefe et al.[28]	1992	• Not further specified	• Not further specified
COOL trial[29]	2011	• **ACS definition** (at least 2 of the following 3 criteria were required): o ischemic chest pain occurring at rest or increasing in frequency and lasting >10 min o ECG changes of ischemia with ST elevation, ST depression or new left bundle branch block o Elevated cardiac biomarkers.	• New focal neurological deficit of vascular origin lasting >24 h. CT scan or MRI imaging results were reviewed for all patients with stroke.
LoDoCo trial[30]	2013	• **MI definition** as per *First Universal definition of Myocardial Infarction*† • *Unstable angina* as evidenced by a recent acceleration of angina unassociated with a rise in serum troponin but angiographic evidence of change in coronary anatomy (as per Braunwald classification types IB and IIB)	• Computed tomography or magnetic resonance imaging proven ischemic stroke judged by the treating neurologist as not being due to atrial fibrillation or intracranial hemorrhage.
Deftereos et al.[31]	2013	• Not further specified	• Not further specified
Giannopoulos et al.[32]	2015	• Not further specified	• Not further specified
Deftereos et al.[33]	2015	• Not further specified	• Not further specified
Zarpelon et al.[34]	2016	• Not further specified	• Not further specified
COLIN trial[35]	2016	• **MI definition** as per *European Society of Cardiology (ESC) Guidelines* for the Management of Acute Myocardial Infarction in patients presenting with ST-segment elevation 2012 (Eur Heart J 2012; 33:2569-619)	• Not further specified

LoDoCo-MI trial[36]	2019	• **MI definition** as *per 3rd Universal Definition of Myocardial Infarction*‡	• Not further specified
COLCOT trial[8]	2019	• **MI** defined according to national guidelines (the exact guidelines had not been defined in the protocol)	• Stroke or TIA (exact definitions were not further specified)
COLCHICINE-PCI trial[16]	2020	• **MI definition** as per *3rd Universal Definition of Myocardial Infarction*‡	• Ischemic stroke, not further specified.
COPS trial[17]	2020	• **ACS definition**: ○ Elevated cardiac troponin ○ ECG changes ○ *STEMI* and *NSTEMI* are defined as ischaemic symptoms, positive troponin and ECG changes. • *Unstable angina* is defined as history consistent with coronary ischaemia with objective evidence of ischaemia (i.e. ECG change) and negative troponin.	• **Non-cardioembolic ischaemic stroke**: CT or MRI-proven ischaemic stroke judged by the treating neurologist as not being due to atrial fibrillation or intracranial haemorrhage.
LoDoCo2 trial[18]	2020	• **MI definition** as per *3rd Universal definition of Myocardial Infarction*‡ • *Unstable angina* as evidenced by a recent acceleration of angina unassociated with a rise in serum troponin but angiographic evidence of change in coronary anatomy (as per Braunwald classification types IB and IIB)	• **Non-cardioembolic ischaemic stroke**: as evidenced by CT or MRI and coded as such by the treating neurologist occurring in the absence of atrial fibrillation, cerebral hemorrhage, lacunae infarction or small vessel disease.

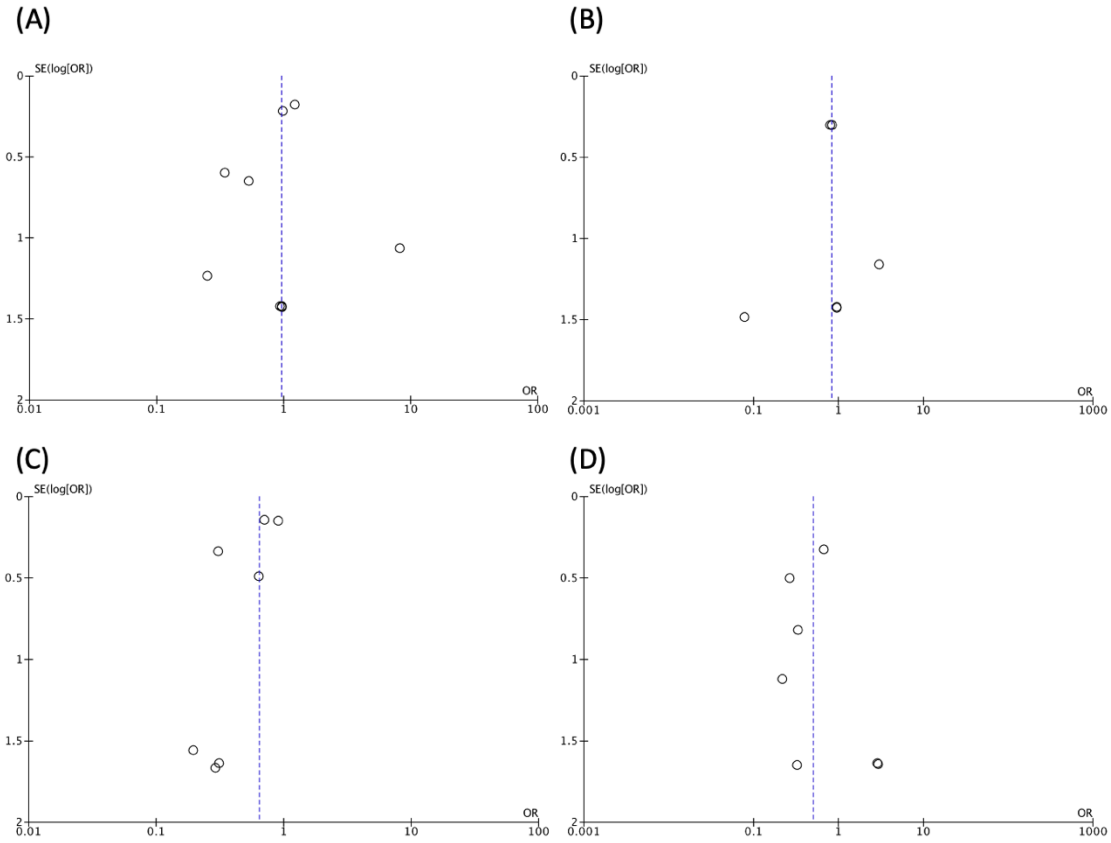
ACS = Acute coronary syndrome; CT = Computed tomography; MRI = magnetic resonance imaging; NSTEMI = Non-ST-segment elevation myocardial infarction; STEMI = ST-segment elevation myocardial infarction; TIA = Transient ischemic attack.

* Of note, some studies only reported their acute coronary syndrome (ACS) definition (e.g. Raju et al.)

† Thygesen K., Alpert J.S., White H.D. and Joint ESC/ACCF/AHA/WHF Task Force for the Redefinition of Myocardial Infarction: "Universal definition of myocardial infarction". J Am Coll Cardiol 2007; **50**: 2173.

‡ Thygesen K., Alpert J.S., Jaffe A.S. and Joint ESC/ACCF/AHA/WHF Task Force for the Redefinition of Myocardial Infarction: "Third universal definition of myocardial infarction". Circulation 2012; **126**: 2020-2035

Figure S1. Funnel plot of logarithmized odds ratio for assessment of publication bias among the included studies for the following outcomes: (A) all-cause mortality; (B) cardiovascular mortality (CV); (C) new myocardial infarction; and (D) stroke / TIA

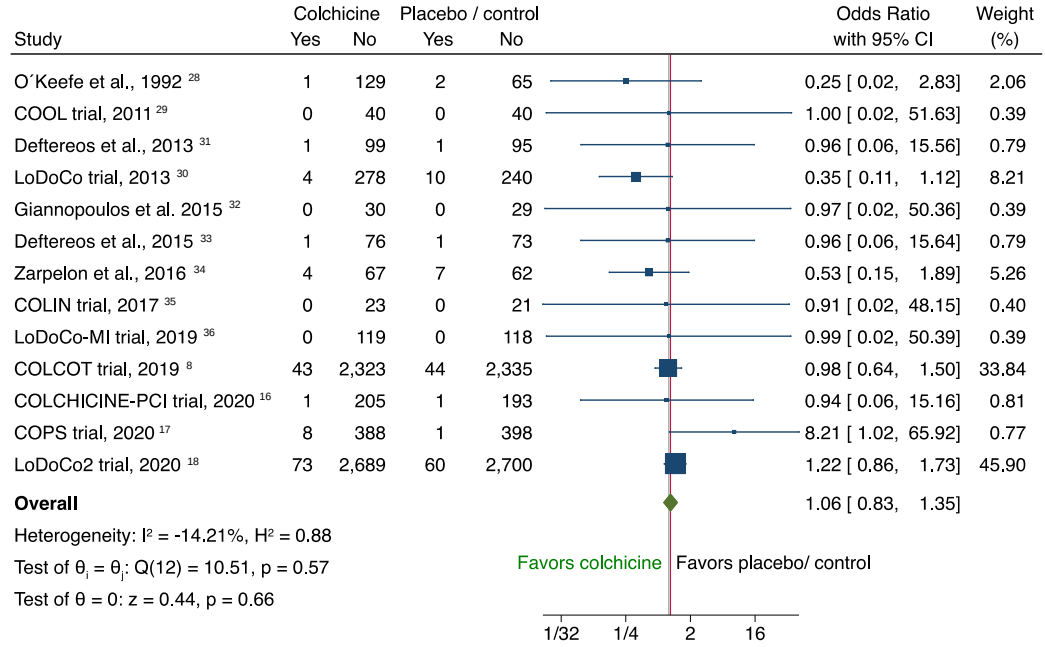


OR = Odds ratio; SE = Standard error.

Figure S2. Colchicine in patients with coronary artery disease (CAD) – Sensitivity analyzes using Mantel-Haenszel (M-H) fixed effect models

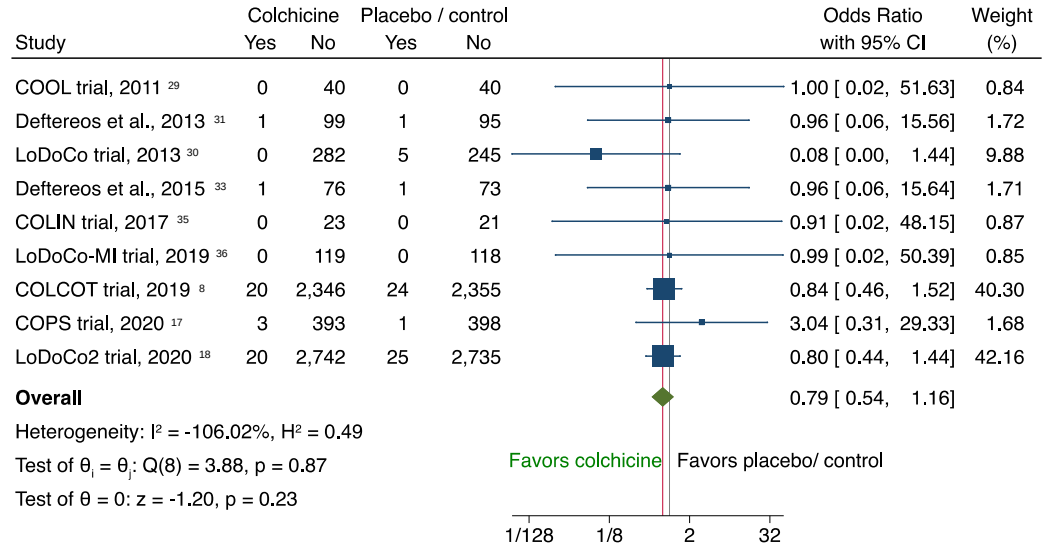
Main outcomes with colchicine compared to placebo / standard therapy in patients with CAD:

(A) All-cause mortality



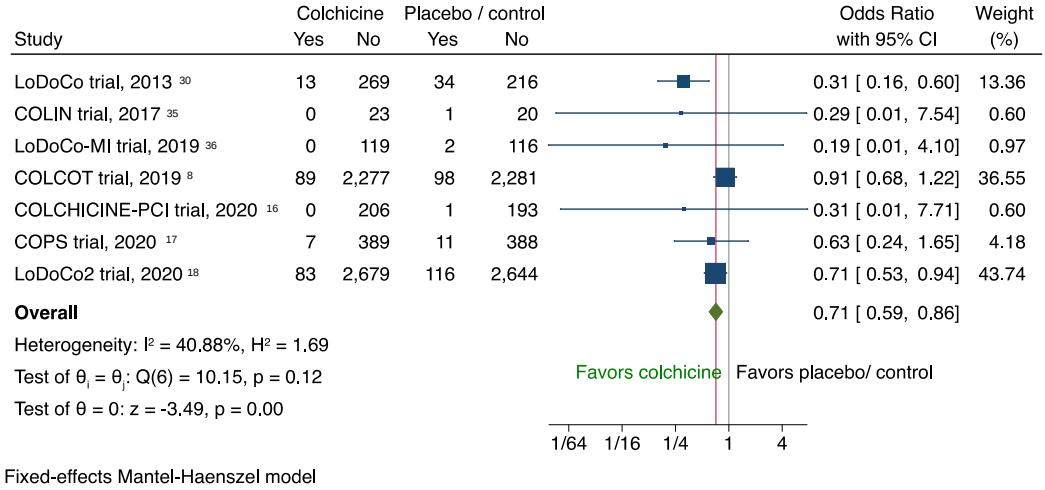
Fixed-effects Mantel-Haenszel model

(B) Cardiovascular (CV) mortality



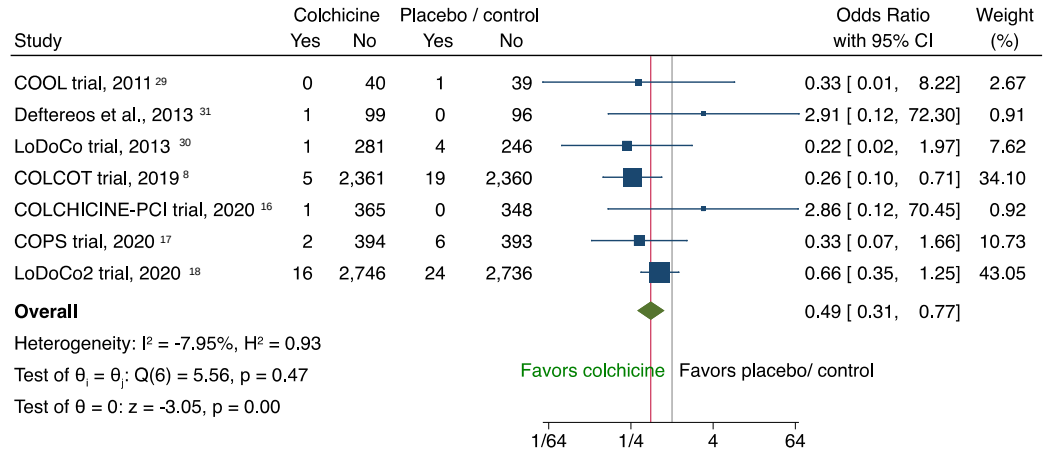
Fixed-effects Mantel-Haenszel model

(C) New myocardial infarction



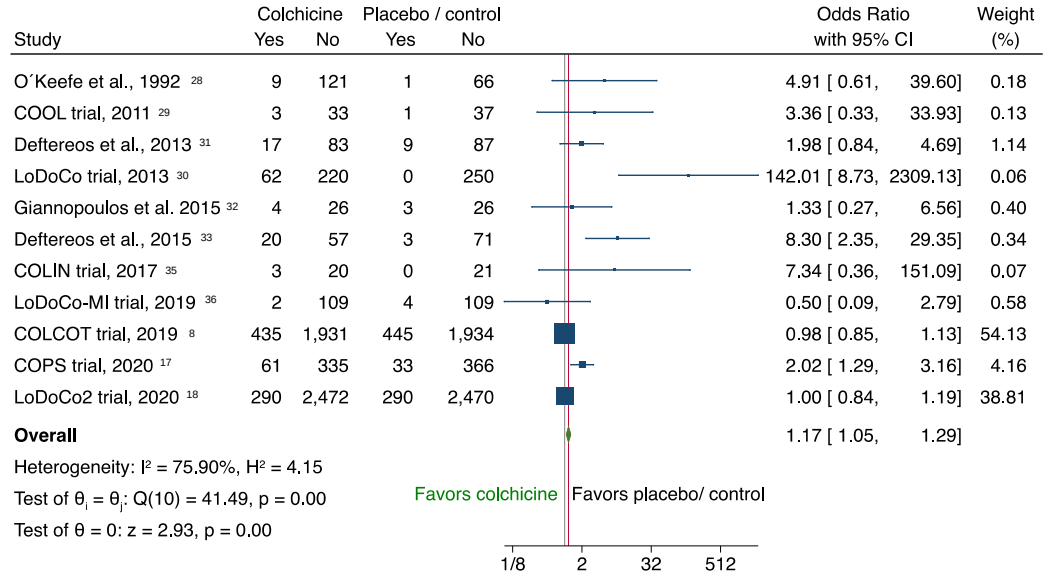
Fixed-effects Mantel-Haenszel model

(D) Stroke or transient ischemic attack (TIA)



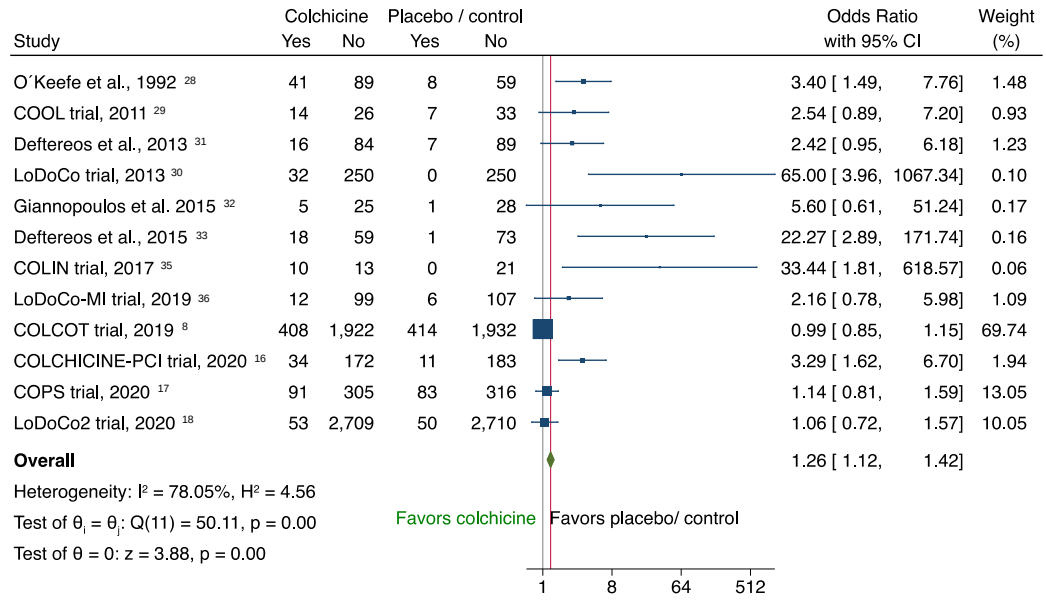
Fixed-effects Mantel-Haenszel model

(E) Therapy discontinuiation with colchicine compared to placebo/ standard therapy



Fixed-effects Mantel-Haenszel model

(F) Gastrointestinal side effects

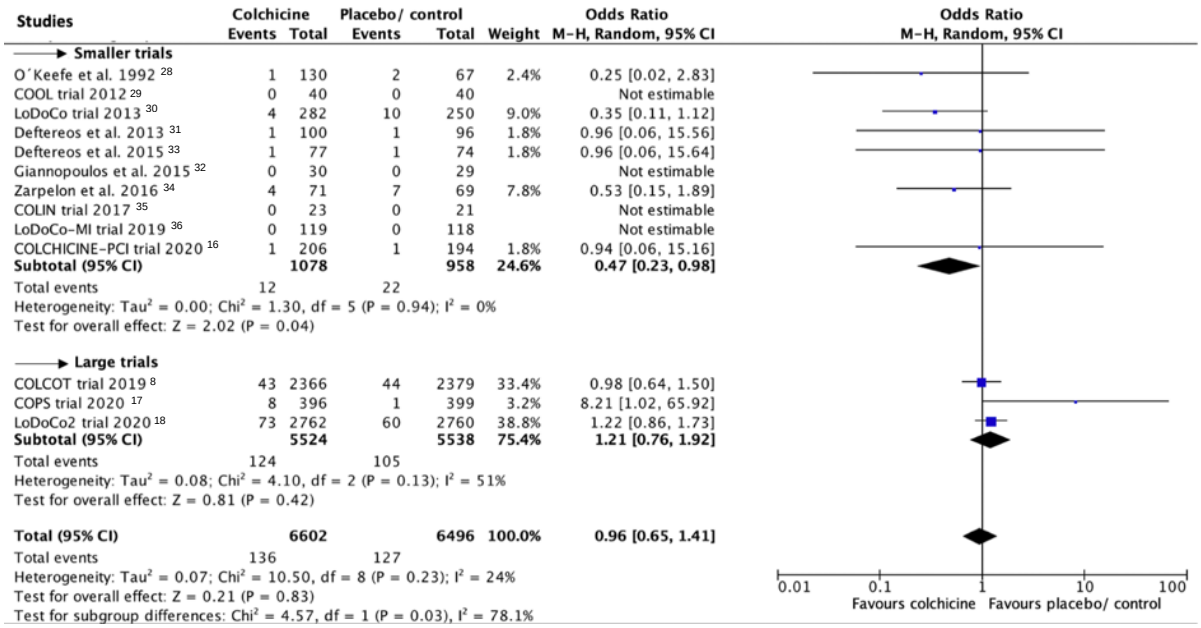


Fixed-effects Mantel-Haenszel model

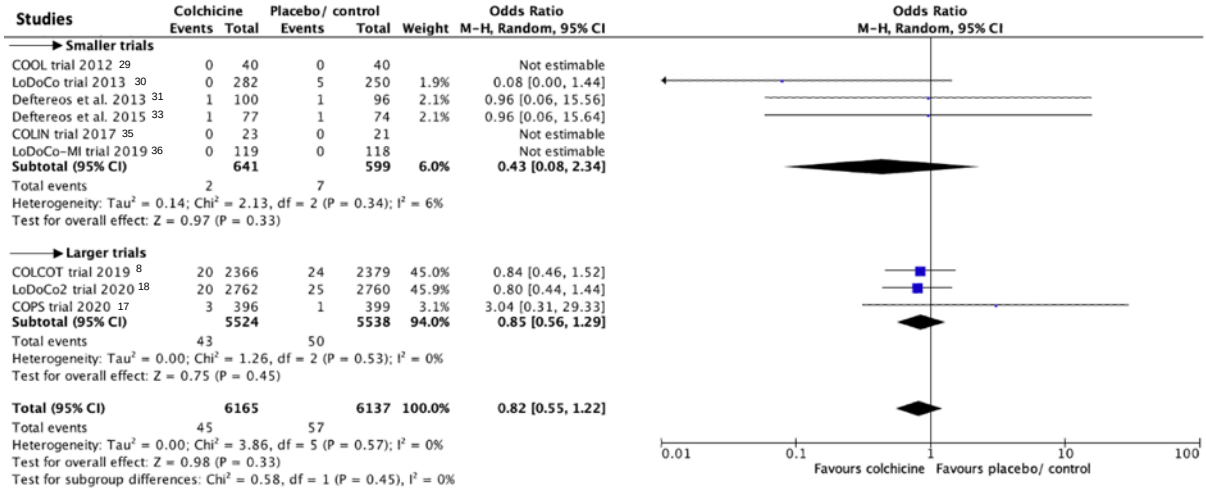
Figure S3. Sensitivity analyzes: Colchicine in patients with coronary artery disease (CAD) – Outcomes in small versus large, randomized trials

Main outcomes with colchicine compared to placebo / standard therapy in patients with CAD:

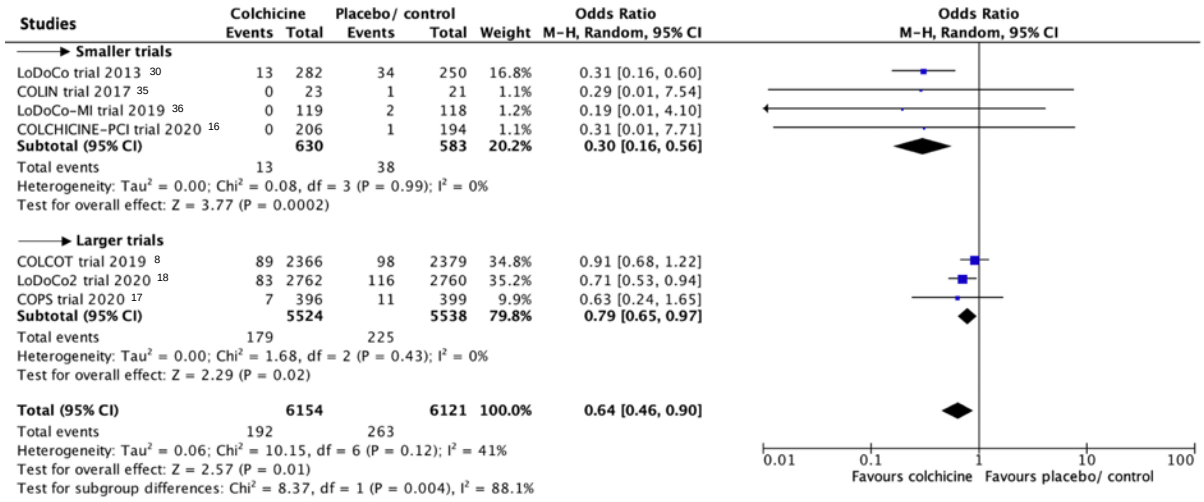
(A) All-cause mortality



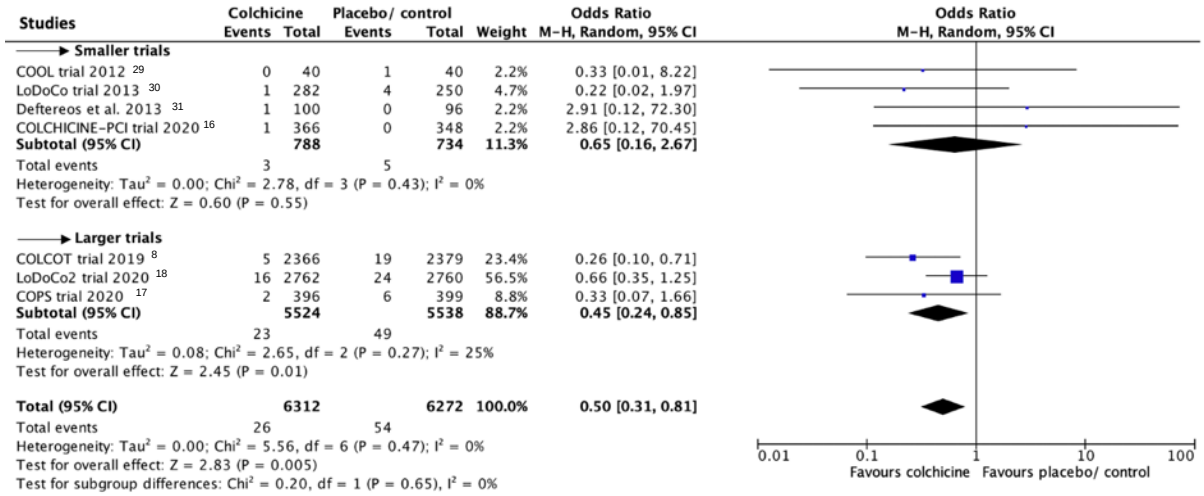
(B) Cardiovascular (CV) mortality



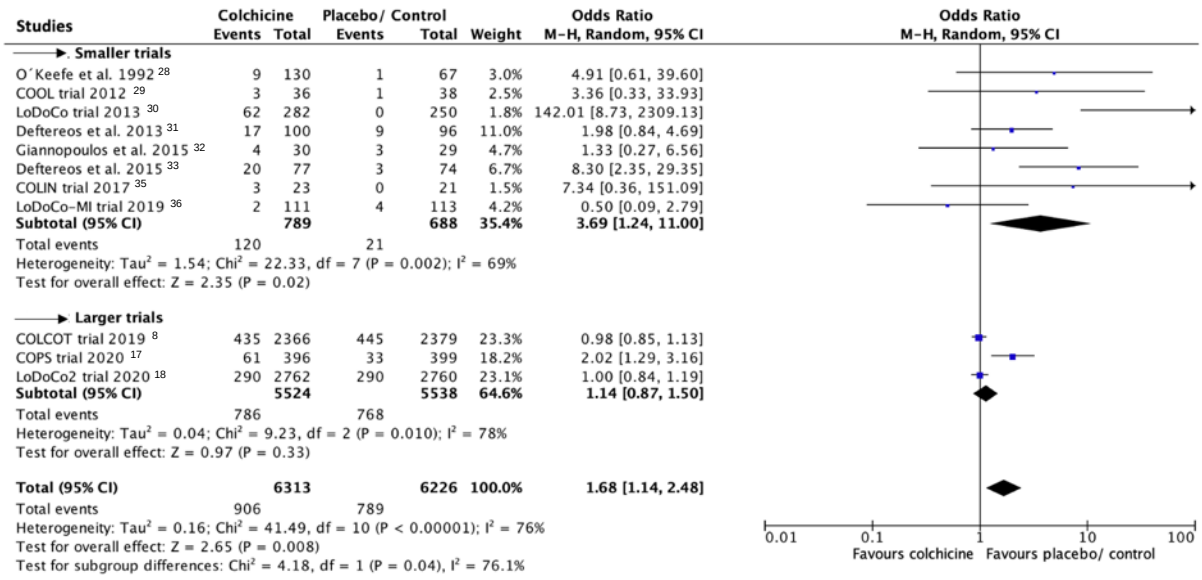
(C) New myocardial infarction



(D) Stroke or transient ischemic attack (TIA)



(E) Therapy discontinuiation with colchicine compared to placebo/ standard therapy



(F) Gastrointestinal side effects

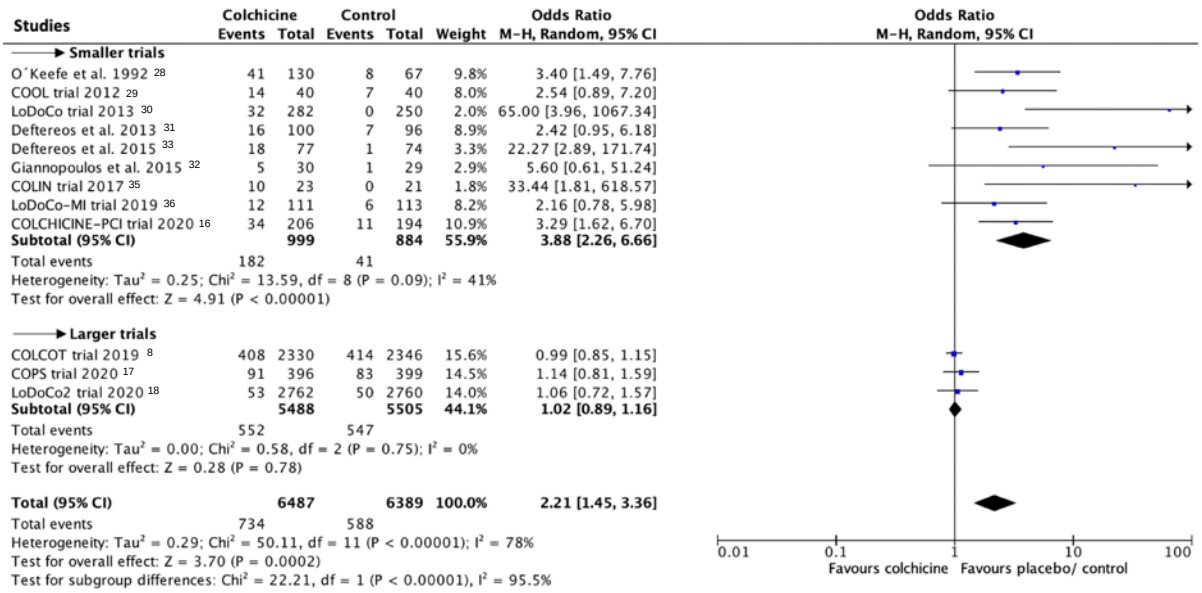
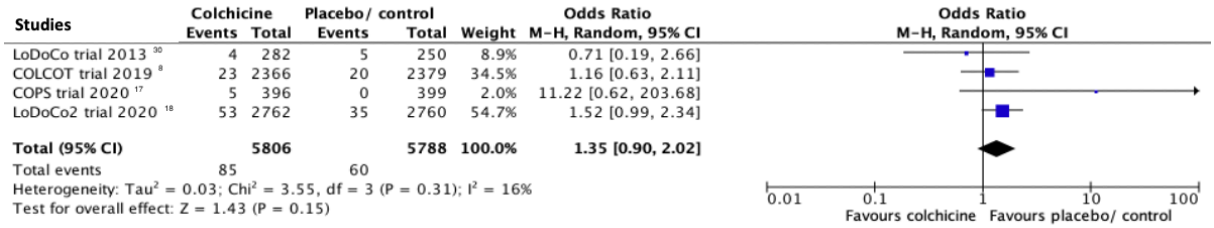
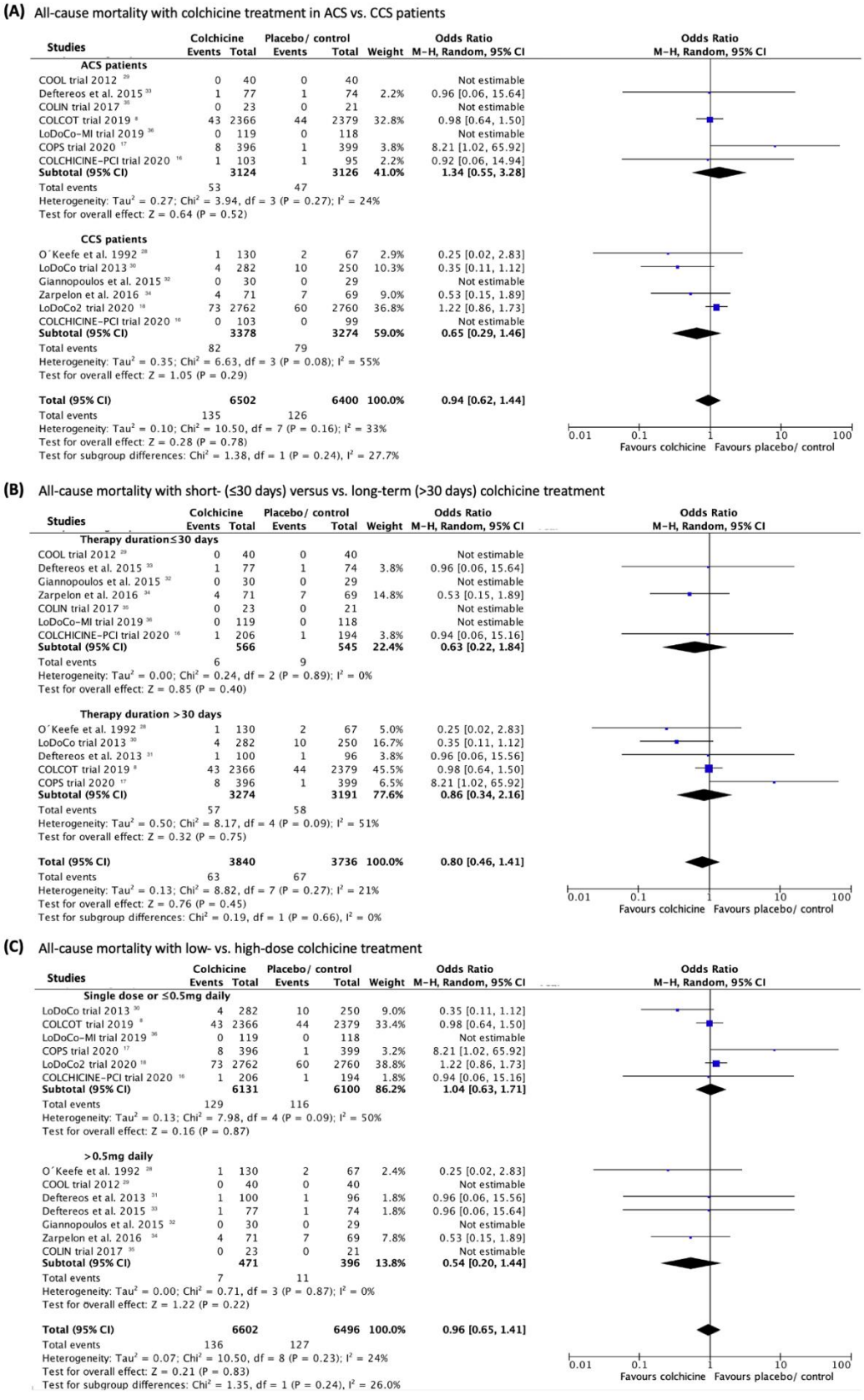


Figure S4. Non-cardiovascular mortality with colchicine compared to placebo/ standard therapy.



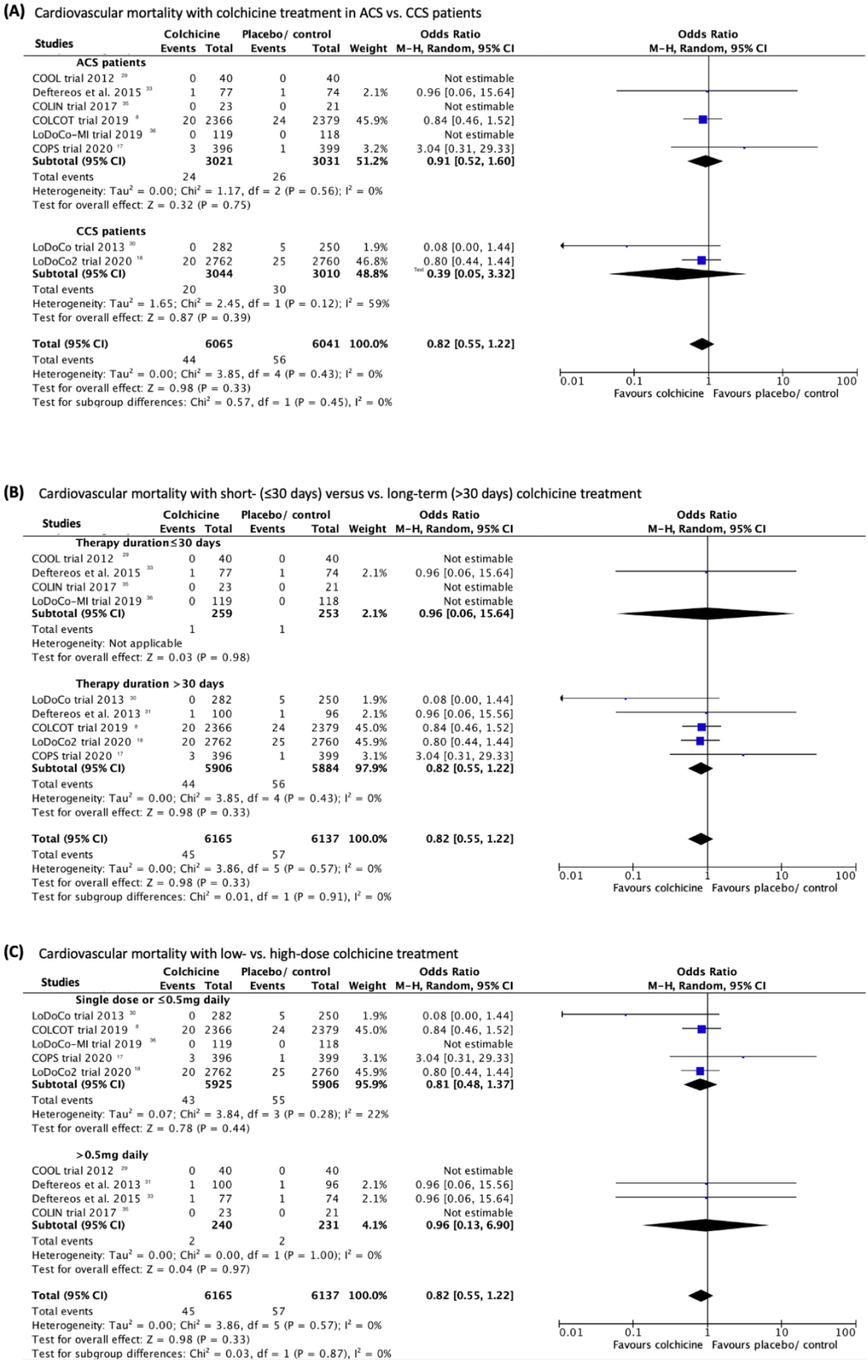
95% CI = 95% confidence interval.

Figure S5. Sub-group analyses assessing all-cause mortality with colchicine versus placebo/ standard therapy in (A) acute versus chronic CAD, (B) for ≤30 days versus >30 days, and (C) at a lower- versus higher-dose regimen.



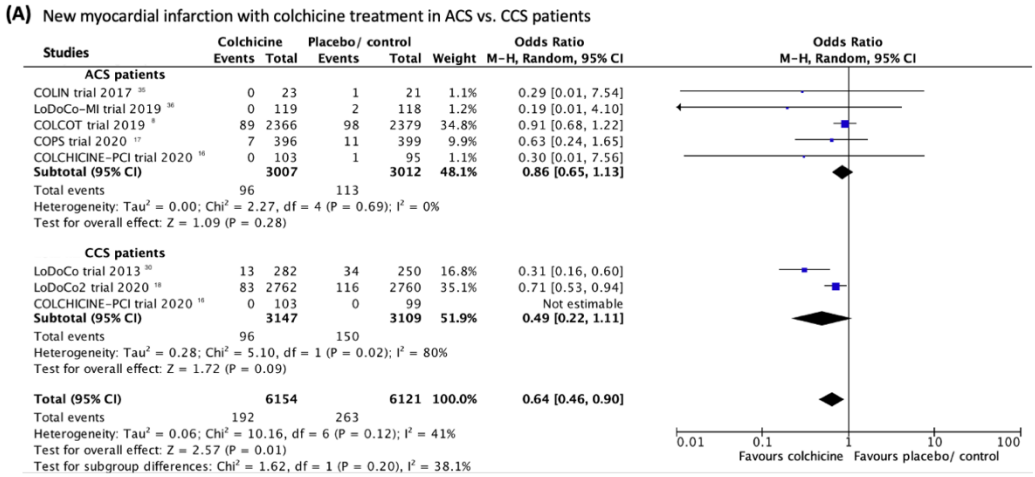
95% CI = 95% confidence interval.

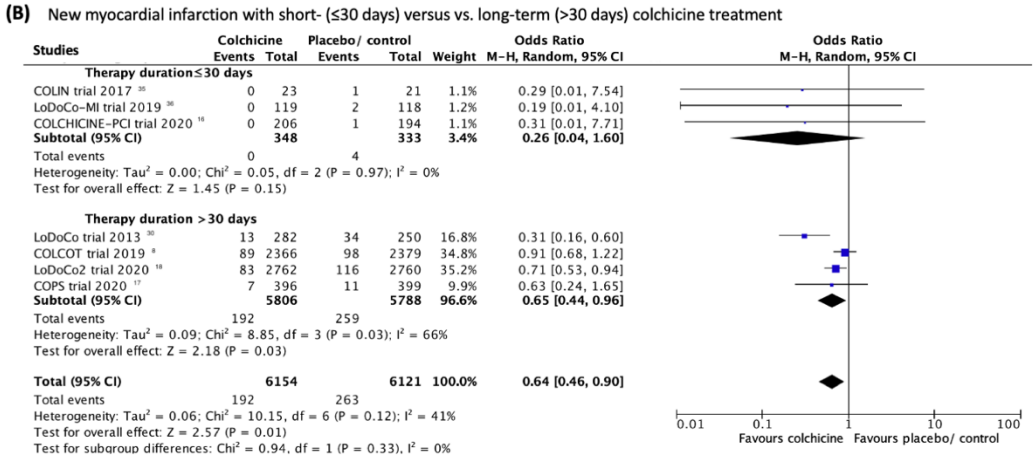
Figure S6. **Sub-group analyses assessing cardiovascular mortality with colchicine versus placebo/ standard therapy in (A) acute versus chronic CAD, (B) for ≤30 days versus >30 days, and (C) at a lower- versus higher-dose regimen**

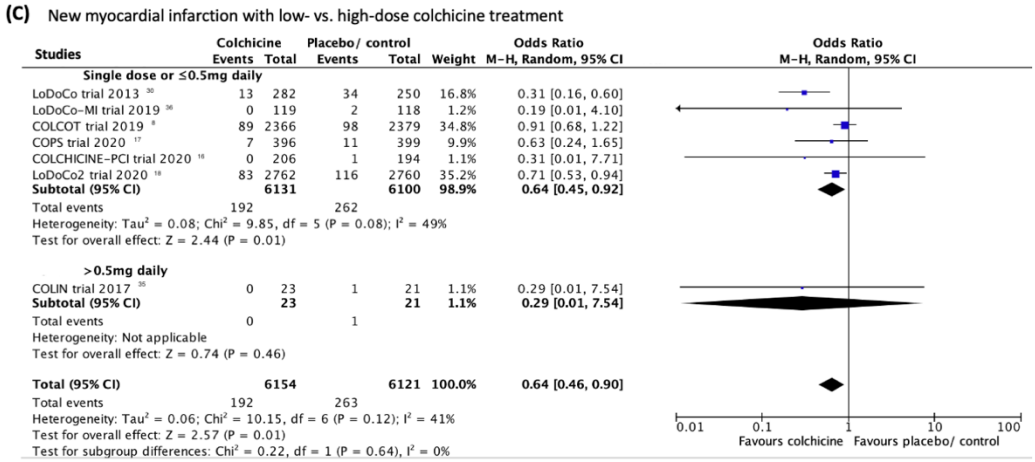


95% CI = 95% confidence interval.

Figure S7. **Sub-group analyses assessing new myocardial infarction with colchicine versus placebo/ standard therapy in (A) acute versus chronic CAD, (B) for ≤30 days versus >30 days, and (C) at a lower- versus higher-dose regimen**

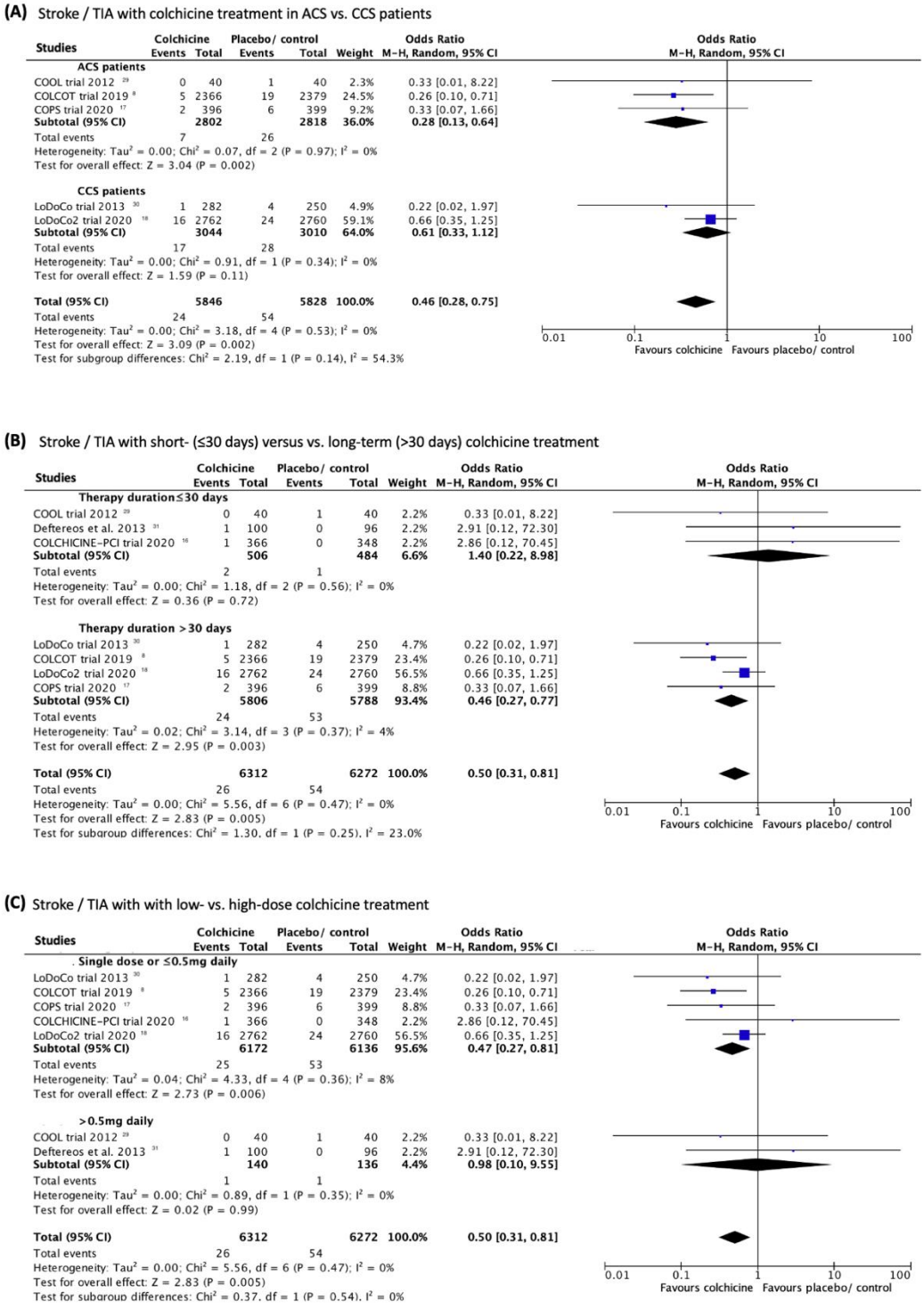






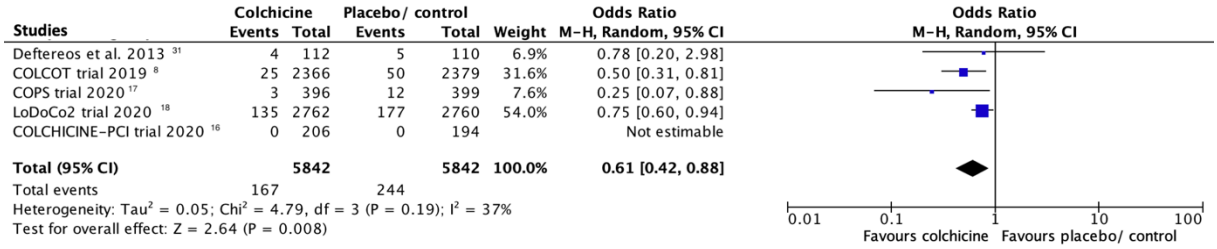
95% CI = 95% confidence interval.

Figure S8 **Sub-group analyses assessing stroke with colchicine versus placebo/ standard therapy in (A) acute versus chronic CAD, (B) for ≤30 days versus >30 days, and (C) at a lower- versus higher-dose regimen**



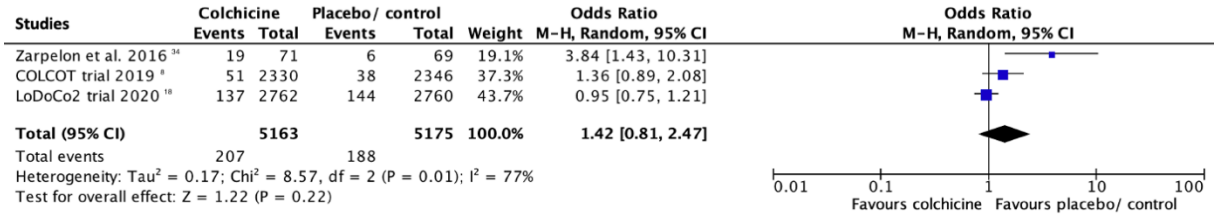
95% CI = 95% confidence interval.

Figure S9. **Ischemia driven or urgent revascularization with colchicine compared to placebo/ standard therapy**



95% CI = 95% confidence interval.

Figure S10. **Infectious complications with colchicine compared to placebo/ standard therapy**



95% CI = 95% confidence interval.